

Phase 3 - Hardening Remote Access

This document covers SSH key-based authentication setup, sshd_config hardening, service state verification, and network identity (hostname/FQDN) configuration performed in Phase 3.

1. SSH Key Generation

An ED25519 key pair was generated and protected with a passphrase, then used to authenticate as the alaa user. Listing of the ~/.ssh directory confirms correct file permissions — the private key (id_ed25519) is readable only by its owner (600), while the public key (id_ed25519.pub) is world-readable, as expected:

```
total 16
-rw-----. 1 lobnaahmed lobnaahmed 464 Jul  4 18:03 id_ed25519
-rw-r--r--. 1 lobnaahmed lobnaahmed  99 Jul  4 18:03 id_ed25519.pub
-rw-----. 1 lobnaahmed lobnaahmed 843 Jul  4 18:11 known_hosts
-rw-----. 1 lobnaahmed lobnaahmed  97 Jul  4 18:11 known_hosts.old
```

```
lobnaahmed@servera:~/rhel-secure-server-project/configs$ ssh-keygen -t ed25519 -C 'alaa@rhel-project'
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/lobnaahmed/.ssh/id_ed25519):
Created directory '/home/lobnaahmed/.ssh'.
Enter passphrase for '/home/lobnaahmed/.ssh/id_ed25519' (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/lobnaahmed/.ssh/id_ed25519
Your public key has been saved in /home/lobnaahmed/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:o0nw5S52V8fWl4JaDD0FTqUMFZRrZeL5ap13isTIVyk alaa@rhel-project
The key's randomart image is:
+--[ED25519 256]--+
|      .o*=o  |
|      *ooo  |
|      . ...*  |
|    o o  o o  |
|    o S .+E.*o|
|    . + oo+ *. |
|    = o.+ * .  |
|    . o . = + ..|
|      . . o..  |
+----[SHA256]-----+
lobnaahmed@servera:~/rhel-secure-server-project/configs$ cat /home/lobnaahmed/.ssh/id_ed25519.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIOeCRenyfI1neQWql/safJkpJGRub103Ch7I3RSKnUHz alaa@rhel-project
```

2. Privilege & Session Activity

The following log excerpt captures privileged session activity around the SSH hardening work: alaa using sudo to run sshd -T (checking the live configuration) and an su session opened by alaa:

```
Jul  4 19:13:46 servera sudo[7529]: pam_unix(sudo:session): session closed for user root
Jul  4 19:14:02 servera sudo[7537]:    alaa : TTY=pts/1 ; PWD=/home/alaa ;
USER=root ; COMMAND=/sbin/sshd -T
Jul  4 19:14:02 servera sudo[7537]: pam_unix(sudo:session): session opened for user root(uid=0) by alaa(uid=1005)
Jul  4 19:14:02 servera sudo[7537]: pam_unix(sudo:session): session closed for user root
Jul  4 19:22:09 servera su[7729]: pam_unix(su-l:session): session opened for user lobnaahmed(uid=1000) by alaa(uid=1005)
```

Note: This excerpt shows privilege usage, not the SSH key-based login itself. The strongest evidence for this phase is the "Accepted publickey for alaa" line from /var/log/secure or journalctl -u sshd, confirmed separately during the project.

```

lobnaahmed@servera:~$ ssh alaa@192.168.136.128
Web console: https://servera.mylab.com:9090/ or https://192.168.136.128:9090/

Register this system with Red Hat Lightspeed: rhc connect

Example:
# rhc connect --activation-key <key> --organization <org>

The rhc client and Red Hat Lightspeed will enable analytics and additional
management capabilities on your system.
View your connected systems at https://console.redhat.com/insights

You can learn more about how to register your system
using rhc at https://red.ht/registration
Last login: Sat Jul  4 18:18:41 2026

```

3. sshd_config Hardening

Password authentication was confirmed disabled and root login blocked, verified live against the running configuration using `sshd -T` (rather than just reading the file, which can differ from what's actually active):

```

passwordauthentication no
permitrootlogin no

```

Note: Key-based login was confirmed working before this change was applied, to avoid a self-inflicted lockout - password authentication was only disabled after verifying the alaa account could authenticate via key.

4. SSH Service State

```

enabled
• sshd.service - OpenSSH server daemon
  Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled; preset: enabled)
  Active: active (running) since Sat 2026-07-04 19:13:11 EEST; 13min ago
  Invocation: 319c8253707c4bb793efe9052c1c31e7
    Docs: man:sshd(8)
          man:sshd_config(5)
  Main PID: 7525 (sshd)
    Tasks: 1 (limit: 10098)
  Memory: 1.3M (peak: 1.5M)
    CPU: 44ms
  CGroup: /system.slice/sshd.service
          └─7525 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"
Jul 04 19:13:11 servera.mylab.com systemd[1]: Starting sshd.service - OpenSSH server daemon...
Jul 04 19:13:11 servera.mylab.com sshd[7525]: Server listening on 0.0.0.0 port 22.
Jul 04 19:13:11 servera.mylab.com sshd[7525]: Server listening on :: port 22.
Jul 04 19:13:11 servera.mylab.com systemd[1]: Started sshd.service - OpenSSH server daemon.

```

The service is both enabled (will start automatically on boot) and actively running, listening on both IPv4 and IPv6 on the default port 22.

5. Hostname / FQDN Configuration

The server's static hostname was set to a proper fully qualified domain name, `servera.rhelproject.local`, via `hostnamectl`:

```
Static hostname: servera.rhelfproject.local
Icon name: computer-vm
Chassis: vm
Machine ID: 4cbd8cbcd7fb443f991ab90abaf25410
Virtualization: vmware
Operating System: Red Hat Enterprise Linux 10.2 (Coughlan)
CPE OS Name: cpe:/o:redhat:enterprise_linux:10.2
Kernel: Linux 6.12.0-211.7.3.el10_2.x86_64
Architecture: x86-64
Hardware Vendor: VMware, Inc.
Hardware Model: VMware20,1
```

Note: Setting the hostname via `hostnamectl` triggered a Polkit authentication prompt rather than a standard `sudo` password check - a distinct privilege-escalation framework used by several modern `systemd` tools, worth understanding separately from `sudo` when troubleshooting permission issues on RHEL.

6. `/etc/hosts`

The default loopback entries were reviewed; no custom entries were required for this single-server lab setup, though the file supports adding friendly hostname mappings for other machines if the environment expands:

```
# Loopback entries; do not change.
# For historical reasons, localhost precedes localhost.localdomain:
127.0.0.1    localhost localhost.localdomain localhost4 localhost4.localdomain4
::1         localhost localhost.localdomain localhost6 localhost6.localdomain6
```